Effat College of Engineering · Computer Science

CS3081 Artificial Intelligence

Assignment: Uncertainty and Bayesian Networks

Instructor: Dr. Naila Marir **Total:** 50 marks (5 exercises × 10 marks)

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Instructions

1. Show all your steps. If you write only the final number, you get at most half of the marks.
2. Write probabilities with 4 decimal places (for example 0.5367), unless the question says something else.
3. Exercises 1 to 4, and Exercise 5 parts (a) and (b), are solved by hand.
4. Exercise 5 part (c) is solved in Python with pgmpy in VS Code, like in the lab.
5. Submit two files on Blackboard: these pages with your answers written in the boxes, scanned as one PDF, and your notebook `assignment_bn.ipynb` with all outputs.
6. This is individual work. You can discuss ideas with classmates, but you must write your own answers.

Exercise 1 — Joint distribution (10 marks)

The Effat shuttle bus is either **on time** or **late**. The weather in Jeddah is either **clear** or **dusty**. The table gives the probability of each pair. For example, the probability that the weather is clear **and** the bus is late is 0.14.

Weather	Shuttle = on time	Shuttle = late
clear	0.56	0.14
dusty	0.12	0.18

1. **(2 marks)** Show that the four numbers in the table make a valid probability distribution.

To have valid distribution, all the individual probabilities must be between 0 and 1, and their sum must equal 1 therefor:

1. 0.56 , 0.14 , 0.12 , 0.18 >= 0
2. 0.56 + 0.14 + 0.12 + 0.18 + = 1

Since all probabilities are non-negative and their sum is exactly 1, it is a valid probability distribution

2. **(2 marks)** Find P(dusty), the probability that the weather is dusty. Then find P(late), the probability that the bus is late.

P(dusty)=P (dusty, on time) +P (dusty, late) =0.12+0.18= 0.3000

P(late)=P (clear, late) +P (dusty, late) =0.14+0.18=0.3200

3. **(2 marks)** Find P (late | dusty) and P (late | clear).

We use the conditional probability formula (P (A | B) = P (A, B)/P(B)

P (late | dusty) = P (late, dusty) / P (dusty) = 0.18 / 0.3 = 0.6000

P (late | clear) = P (late, clear) / P (clear) = 0.14 // 0.7 = 0.2000

4. **(2 marks)** Find P (dusty | late). Is it the same as P (late | dusty)? Explain your answer in one sentence.

P (dusty | late) = P (dusty, late) / P(late) = 0.18 /0.32 = 0.5625

P (late | dusty) = P (late, dusty) / P (dusty) = 0.18 / 0.3 = 0.6000

No, P (dusty | late) is not the same as P (late | dusty) because they condition on different sample spaces , P (dusty) doesn't equal, P (late)

5. **(2 marks)** Are the weather and the bus independent? Use numbers to explain your answer.

Two variables are independent if and only if P (A, B) = P(A) * P(B) for all outcomes.
Let's test (clear, late):
- Product: P(clear) times P(late) = 0.7000 * 0.3200 = 0.2240
- Actual joint probability from table: P (clear, late) = 0.1400

Since 0.1400 not equal to 0.2240, the weather and the bus are not independent.

Exercise 2 — Bayes' rule: a spam filter (10 marks)

An email program wants to find spam emails. It knows these facts:

- 30% of all emails are spam. So P(spam) = 0.3 and P(normal) = 0.7.
- In spam emails, the word "free" appears 60% of the time: P(free | spam) = 0.6.
- In normal emails, the word "free" appears 5% of the time: P(free | normal) = 0.05.

1. **(3 marks)** Find P(free), the probability that any email contains the word "free". Write the name of the rule you use.

Use the law of total probability:

P(free) = P (free | spam) P(spam)+P (free | normal) P(normal)

P(free) = (0.6 * 0.3) + (0.05 * 0.7)

P(free) = 0.1800+0.0350=0.2150

2. **(3 marks)** A new email contains the word "free". Use Bayes' rule to find P(spam | free).

Using the Bayes' Rule:

P (spam | free) = P (free | spam) P (spam) / P (free)

P (spam | free) = 0.6*0.3 / 0.2150 = 0.8372

3. **(3 marks)** The program also checks the word "winner": P (winner | spam) = 0.4 and P (winner | normal) = 0.02. Assume that, once we know if an email is spam or normal, the two words appear independently of each other. A new email contains **both** "free" and "winner". Find P (spam | free, winner).

For Spam: P (free, winner | spam) = 0.6 *0.4 = 0.2400
For Normal: P (free, winner | normal) = 0.05 *0.02 = 0.0010

p (free, winter) = (0.2400 * 0.3) + (0.0010 * 0.7) = 0.0727

P (spam | free, winner) = 0.0720 / 0.0727 = 0.9904

4. **(1 mark)** Compare your answers to parts 2 and 3. Did the probability of spam go up or down when we added the word "winner"? Explain why in one sentence.

The probability of spam went up (from 0.8372 to 0.9904) because the presence of the word "winner" provides additional strong evidence since it is much more likely to appear in spam emails than normal ones.

Exercise 3 — Bayesian network: structure and joint probability (10 marks)

In Jeddah summers, a **heatwave** (H) can cause an **AC overload** (A) in a campus building. A heatwave and an AC overload can both cause a **power cut** (C). A power cut can cause the **lab session to be cancelled** (L). Each variable is yes or no.

The network has 4 arrows: H → A, H → C, A → C, and C → L. The tables give only the probability of "yes". The probability of "no" is 1 minus the probability of "yes".

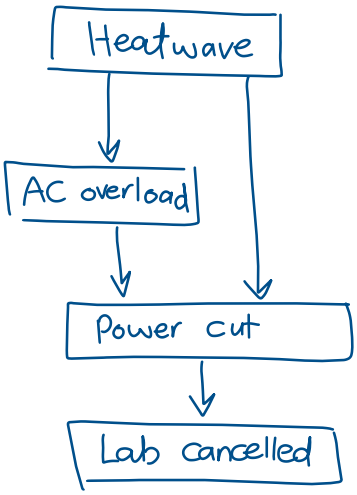
P (H = yes)
0.3

H	P (A =
yes	0
no	0

H	A	P (C = A)
yes	yes	
yes	no	
no	yes	
no	no	

C	P (L = yes | C)
yes	0.8
no	0.1

1. **(2 marks)** Draw the network. For each node, write its parents.



2. **(2 marks)** Write P (H, A, C, L) as a product of the probabilities from the network's tables.

P (H, A, C, L) = P (H) * P (A | H) * P (C | H, A) * P (L | C)

3. **(2 marks)** A full joint table for 4 yes/no variables need 15 numbers ($2^4 - 1$). How many numbers do the 4 tables of this network need? Count only the "yes" probabilities, and show how you counted.

To count the required "yes" probabilities from the tables:
Table P (H): **1** number (2^0)
Table P (A | H): 2 (2^1)
Table P (C | H, A) 4 (2^2)
Table P (L | C) 2 (2^1)
Total = 1 + 2 + 4 + 2 = 9 numbers

4. **(2 marks)** Find P (H = yes, A = no, C = yes, L = yes).

P (H = yes) = 0.3
P (A = no | H = yes) = 1 – 0.6 = 0.4
P (C = yes | H = yes, A = no) = 0.2
P (L= yes | C = yes) = 0.8
P (yes, no, yes, yes) = 0.3*0.4*0.2*0.8 = 0.0192

5. **(2 marks)** Find P (H = yes | C = yes). Then give P (H = yes | C = yes, L = yes) without doing a new calculation, and explain why the two answers are the same.

Due to the network structure, (L) is conditionally independent of (H) given its parent (C), therefore:

P (H = yes | C = yes, L = yes) = P (H = yes | C = yes).

The two answers are identical because once the status of the power cut (C) is known, any downstream effect like the lab cancellation (L) provides no new information about the upstream heatwave (H)

Exercise 4 — Inference by enumeration, challenge question (10 marks)

This is the **challenge question** of the assignment. Show every step.

Use the network and tables from Exercise 3. Today the lab session **was cancelled** (L = yes).

1. **(4 marks)** Use inference by enumeration to find P (H = yes | L = yes), the probability that there was a heatwave. Write the query variable, the evidence and the hidden variables, then show all sums and the normalization.

P (H = yes | L = yes)
Query variable: H
Evidence variable: L = yes
Hidden variables: A, C

We compute the full joint distribution by summing over all hidden variables:
P (H, L = yes) = $\sum_A \sum_C P(H)P(A \mid H)P(C \mid H, A)P(L = \text{yes} \mid C)$

Case 1: For H = yes
- If (A =yes): 0.3*0.6*[(0.7 * 0.8) + (0.3 * 0.1)] = 0.18 * [0.56 + 0.03] = 0.1062
- If (A = no): 0.3* 0.4* [(0.2* 0.8) + (0.8* 0.1)] = 0.12* [0.16 + 0.08] = 0.0288
- Total P (H =yes, L =yes) = 0.1062 + 0.0288 = **0.1350**

Case 2: For H = no
- If (A =yes): 0.7* 0.1*[(0.5* 0.8) + (0.5* 0.1)] = 0.07* [0.40 + 0.05] = 0.0315
- If (A = no): 0.7*0.9*[(0.05*0.8) + (0.95* 0.1)] = 0.63* [0.04 + 0.095] = 0.08505
- Total P (H = no, L = yes) = 0.0315 + 0.08505 = **0.11655**

Normalization:

alpha =1/0.1350 + 0.11655= 1/0.25155
P (H=yes | L=yes) =0.1350 / 0.25155= **0.5367**

2. **(3 marks)** Now find P (A = yes | L = yes), the probability that there was an AC overload. Which variables are hidden this time?

P (A = yes | L = yes)

Hidden variables this time: H, C

P (A = yes, L = yes) = 0.1062 (from H=yes) + 0.0315 (from H=no) = 0.1377

P (A = no, L =yes) = 0.0288 (from H=yes) + 0.08505 (from H=no) = 0.11385

P (A=yes, L=yes) = 0.1377/0.1377+0.11385=0.1377/0.25155= 0.5474

3. **(3 marks)** A technician checks the building and reports that there was **no** AC overload (A = no). The lab is still cancelled (L = yes). Find P (H = yes | L = yes, A = no). Compare it with your answer to part 1: did the probability of a heatwave go up or down? Use the network to explain why in two or three sentences.

$$P(H=\text{yes} \mid L=\text{yes}, A=\text{no}) = \frac{P(H=\text{yes}, A=\text{no}, L=\text{yes})}{P(A=\text{no}, L=\text{yes})} = \frac{0.0288}{0.11385} \approx 0.2530$$

Comparison & explanation:
The probability of a heatwave went down significantly (from 0.5367 to 0.2530). This is due to intercausal reasoning: a heatwave normally causes an AC overload, which sequentially causes power cuts. Knowing explicitly that there were no AC overload blocks one of the major causal pathways through which a heatwave leads to a lab cancellation, making the heatwave less likely to be the cause.

Exercise 5 — Approximate inference by sampling (10 marks)

Now we find the same probability, P (H = yes | L = yes), by **sampling**. Sampling gives an estimate, not the exact answer.

(a) Rejection sampling (3 marks). Reminder: in rejection sampling, we keep only the samples that agree with the evidence (here L = yes), and we throw away the others. Below are 12 samples from the network of Exercise 3.

#	H	A	C	L
1	no	no	no	no
2	yes	yes	yes	yes
3	no	no	no	no
4	no	yes	yes	yes
5	yes	yes	no	no
6	no	no	no	yes
7	yes	no	yes	yes
8	no	no	no	no
9	no	no	no	no
10	yes	yes	yes	no
11	no	no	no	no

12	no	no	yes	yes

1. Write the numbers of the samples we keep and the samples we throw away. Then estimate P (H = yes | L = yes).

Samples we keep: 2, 4, 6, 7, 12 (Total: 5 samples match the evidence (L = yes)
Samples we throw away: 1, 3, 5, 8, 9, 10, 11 (Total: 7 samples)

Estimate: Out of the 5 kept samples H =yes occurs in samples 2 and 7.
P (H = yes | L = yes) = 2/5 = 0.4000

2. What fraction of the 12 samples did we throw away? Why is this a problem for rejection sampling?

Fraction thrown away: 7/12=0.5833, this is a problem because if evidence is rare, rejection sampling wastes computing power throwing away the majority of data, requiring too many samples to achieve accuracy.

(b) Likelihood weighting (3 marks). Reminder: in likelihood weighting, we do not sample the evidence. We fix it (here L = yes), we sample the other variables, and we give each sample a **weight**. The weight is the probability of the evidence given the sample's values: here, P (L = yes | C). Below are 6 samples.

#	H	A	C	L
1	no	no	no	yes (fixed)
2	yes	yes	yes	yes (fixed)
3	no	no	no	yes (fixed)
4	yes	no	no	yes (fixed)
5	no	yes	yes	yes (fixed)
6	yes	yes	yes	yes (fixed)

1. Write the weight of each sample.

Weights W= P (L = yes | C):
Sample 1: C = no = 0.1000
Sample 2: C = yes = 0.8000
Sample 3: C = no = 0.1000
Sample 4: C = no = 0.1000
Sample 5: C = yes = 0.8000
Sample 6: C = yes = 0.8000

2. Estimate P (H = yes | L = yes) using the weights: add the weights of the samples where H = yes, then divide by the sum of all weights.

Sum of weights where H =yes (Samples 2, 4, 6) = 0.8 + 0.1 + 0.8 = 1.7

Total sum of all weights = 0.1 + 0.8 + 0.1 + 0.1 + 0.8 + 0.8 = 2.7
P (H= yes | L= yes) =1.7/2.7 =0.6296

3. Compare your estimates from (a) and (b) with the exact answer from Exercise 4. Why are they different? How can we make them closer to the exact answer?

The estimates (0.4000 and 0.6296) differ from the exact answer (0.5367) because sampling methods rely on random configurations, and our sample sizes (12 and 6) are extremely small. We can make them closer by generating a significantly higher number of samples like 10,000+

(c) Python with pgmpy (4 marks). Answer this part in your notebook, not on paper. In VS Code, create `assignment_bn.ipynb`.

1. Build the network of Exercise 3 with `DiscreteBayesianNetwork` and `TabularCPD`, like in the lab. Name the nodes `heatwave`, `overload`, `powercut` and `cancel`, and use the values `yes` and `no`. Print `model.check_model()`. It must print `True`.
2. Use `VariableElimination` to find P(H | L = yes). Your answer must be the same as in Exercise 4.
3. Use rejection sampling with 10,000 samples to estimate P(H = yes | L = yes). Hint: import `BayesianModelSampling` from `pgmpy.sampling` and `State` from `pgmpy.factors.discrete`. Then use `BayesianModelSampling(model).rejection_sample(evidence=[State("cancel", "yes")], size=10000)`. This gives a table of samples. Count the fraction of rows where `heatwave` is `yes`.
4. In a Markdown cell, write two sentences that compare the exact answer and the sampling estimate.